A PROJECT REPORT

On

S K Bike Point -Two-Wheeler Workshop & Customer Management System

Submitted by

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in partial fulfilment for the award of the degree of

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IN

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under the guidance of

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DECLARATION

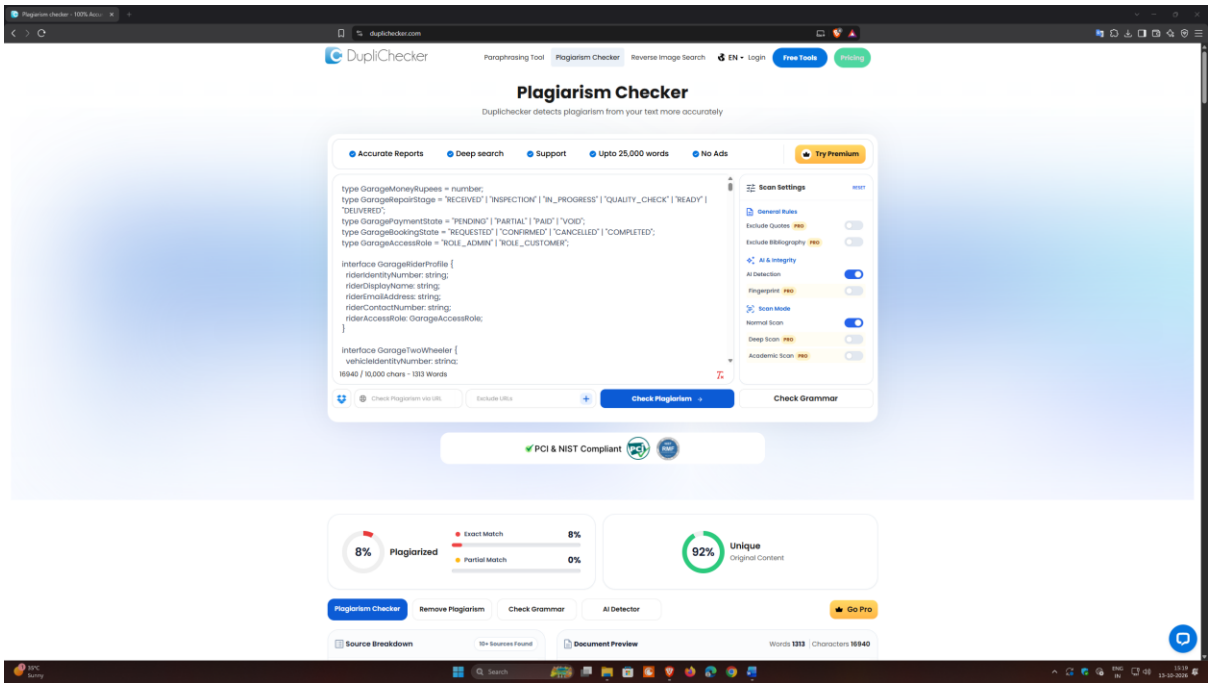
I, **Sumeet Jitendra Yadav**, hereby declare that the project entitled **"S K Bike Point -Two Wheeler Workshop & Customer Management System"** submitted in the partial fulfilment for the award of Bachelor of Science in Computer Science during the academic year 202- – 202- is my original work and the project has not formed the basis for the award of any degree, associateship, fellowship or any other similar titles.

Signature of the Student:

Place: Mumbai

Date

PLAGIARISM REPORT



PREFACE

Two-wheeler repair and maintenance services in suburban workshops are traditionally operated through handwritten job cards, verbal estimates, paper registers, and offline communication. Bike owners often face lack of transparency regarding repair turnaround times, component replacements, and service billing before receiving their vehicles. For an automobile workshop, managing active repair stages, floor mechanic assignments, spare parts inventory, and GST compliance across separate manual ledgers is cumbersome, error-prone, and time-consuming.

The S K BIKE POINT project addresses this gap through a modern cloud-native digital workshop management and customer information platform developed for S K BIKE POINT. The platform provides customers with an organized and interactive way to track real-time repair progress via Zero-Login Vehicle Tracking (using only vehicle registration numbers), explore service maintenance packages, view itemized spare parts diagnostic estimates, book service appointments online, and download GST tax invoices. It also provides a secure administrative dashboard through which workshop managers can oversee active floor job cards, allocate technicians, monitor inventory thresholds, and track daily workshop revenue. The system is implemented using a 3-Tier Layered Architecture with React 18, TypeScript, and Tailwind CSS on the frontend, Java 21 and Spring Boot 3.3 REST API on the backend, and cloud-hosted PostgreSQL 16 on Supabase with HikariCP connection pooling, JWT authentication, and BCrypt security.

This report documents the system analysis, requirements, system design, database structure, architecture, implementation, testing, deployment, and conclusion of the S K BIKE POINT platform. The documentation includes the major UML and system-design diagrams, including the Class Diagram, Use Case Diagram, System Flow Chart, 3-Tier Architecture, Sequence Diagram, ER Diagram, and Deployment Diagram.

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I also acknowledge the open-source community and the developers behind Java 21, Spring Boot 3, React 18, TypeScript, Tailwind CSS, PostgreSQL 16, Supabase, JJWT, BCrypt, Docker, and Render, whose robust frameworks, libraries, and documentation contributed significantly to the successful development of this project.

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ABSTRACT

Two-wheeler maintenance and garage management services are often hindered by manual paperwork, untracked spare parts leakage, lack of live repair visibility, and fragmented customer records, making it difficult for bike owners to monitor service progress and verify replaced components. This project presents S K BIKE POINT, a cloud-native full-stack web platform engineered to provide a centralized, transparent, and automated digital ecosystem for two-wheeler motorcycle and scooter service workshops.

The platform allows motorcycle and scooter owners to track live 5-stage mechanical repair progress zero-login using their registration number, schedule service appointments online, review diagnostic notes recorded by master mechanics, and access complete vehicle maintenance histories with instant 18% GST invoice generation. A secure administrative command center enables the workshop proprietor and service advisors to create digital job cards, requisition spare parts with automatic inventory stock decrementing, monitor low-stock consumable alert thresholds, and analyze real-time business metrics including daily gross revenue and job completion rates.

The platform is built using React 18 and TypeScript with Tailwind CSS on the frontend and Java 21 with Spring Boot 3.3 on the backend, utilizing a cloud PostgreSQL 16 relational database hosted on Supabase (AWS ap-south-1 Mumbai) with HikariCP connection pooling over encrypted TLS. Stateless JWT authentication and BCrypt password hashing enforce role-based access control (ROLE_ADMIN and ROLE_CUSTOMER), while CORS configuration, input sanitization, and structured exception handlers protect application endpoints. The system was designed using a 3-Tier Layered Architecture and documented through UML models including the Domain Class Diagram, Use Case Diagram, System Flowchart, Sequence Diagrams, Entity-Relationship (ER) Diagrams, and Cloud Deployment Architecture

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